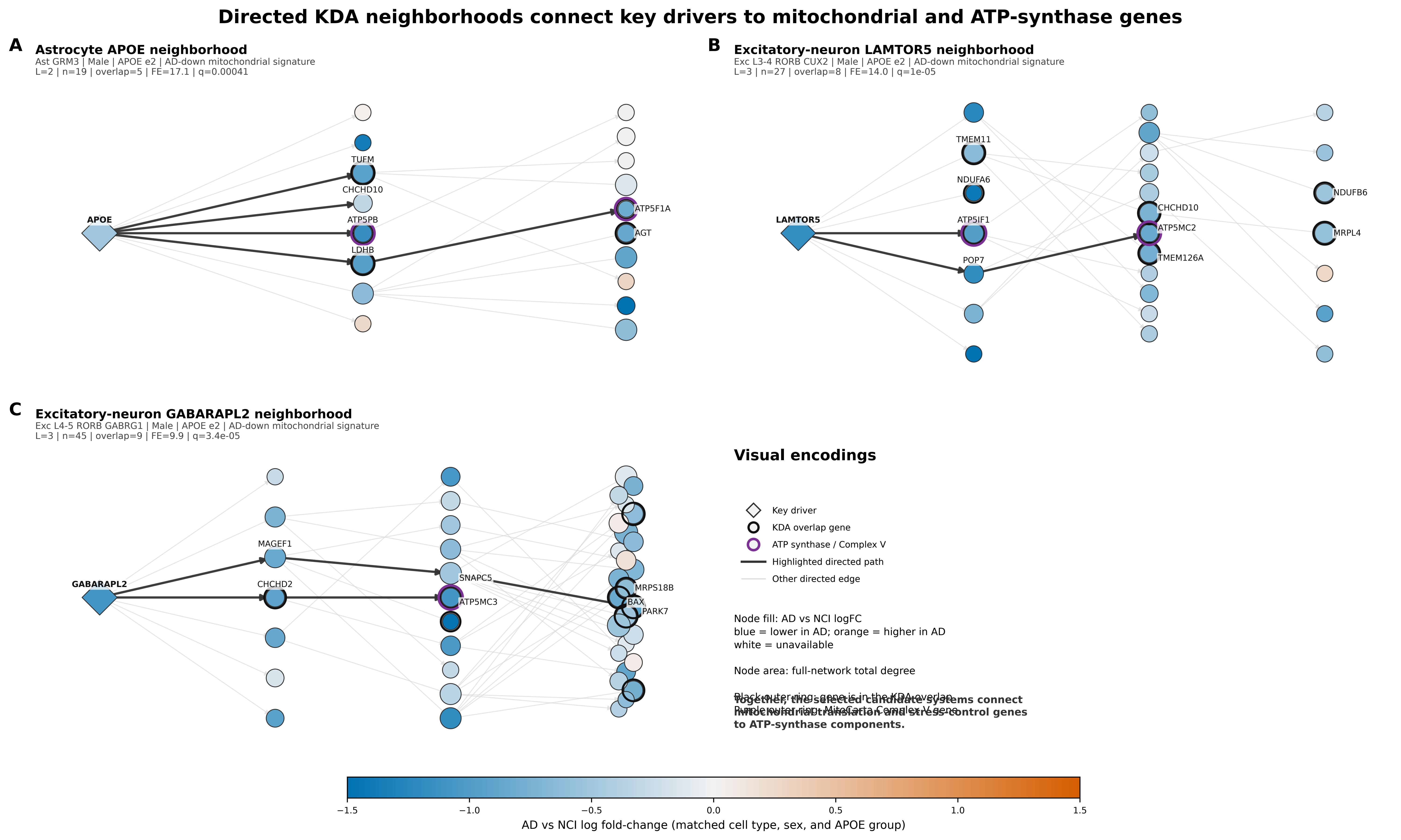
Directed KDA neighborhoods connect key drivers to mitochondrial and ATP-synthase genes
A
Astrocyte APOE neighborhood
Ast GRM3 | Male | APOE e2 | AD-down mitochondrial signature
L=2 | n=19 | overlap=5 | FE=17.1 | q=0.00041
APOE
TUFM
CHCHD10
ATP5PB
LDHB
ATP5F1A
AGT
B
Excitatory-neuron LAMTOR5 neighborhood
Exc L3-4 RORB CUX2 | Male | APOE e2 | AD-down mitochondrial signature
L=3 | n=27 | overlap=8 | FE=14.0 | q=1e-05
LAMTOR5
TMEM11
NDUFA6
ATP5IF1
POP7
CHCHD10
ATP5MC2
TMEM126A
NDUFB6
MRPL4
C
Excitatory-neuron GABARAPL2 neighborhood
Exc L4-5 RORB GABRG1 | Male | APOE e2 | AD-down mitochondrial signature
L=3 | n=45 | overlap=9 | FE=9.9 | q=3.4e-05
GABARAPL2
MAGEF1
CHCHD2
SNAPC5
ATP5MC3
MRPS18B
BAX
PARK7
Visual encodings
Key driver
KDA overlap gene
ATP synthase / Complex V
Highlighted directed path
Other directed edge
Node fill: AD vs NCI logFC
blue = lower in AD; orange = higher in AD
white = unavailable
Node area: full-network total degree
Black outer ring: gene is in the KDA overlap
Purple outer ring: MitoCarta Complex V gene
Together, the selected candidate systems connect
mitochondrial translation and stress-control genes
to ATP-synthase components.
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
AD vs NCI log fold-change (matched cell type, sex, and APOE group)